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9.Compare Linear and Polynomial Regression using Python

1. House Price Dataset

Dataset: house.csv

Area	Price
500	25
800	40
1000	50
1200	60
1500	75

PROGRAM

```
import pandas as pd

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

df = pd.read_csv("house.csv")
X = df[["Area"]]
y = df["Price"]
linear = LinearRegression().fit(X, y)
poly = PolynomialFeatures(degree=2)
Xp = poly.fit_transform(X)
polynomial = LinearRegression().fit(Xp, y)
sample = [[1100]]
print("Linear:", round(linear.predict(sample)[0], 2))
print("Polynomial:", round(
    polynomial.predict(poly.transform(sample))[0], 2))
```

OUTPUT

Linear: 55.0

Polynomial: 55.18

2. Salary Prediction Dataset

Dataset: salary.csv

Experience	Salary
1	3
2	4
3	5.5
4	7
5	9

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
df = pd.read_csv("salary.csv")
X = df[["Experience"]]
y = df["Salary"]
lr = LinearRegression().fit(X, y)
p = PolynomialFeatures(degree=2)
Xp = p.fit_transform(X)
pr = LinearRegression().fit(Xp, y)
sample = [[6]]
print("Linear:", round(lr.predict(sample)[0], 2))
print("Polynomial:", round(
    pr.predict(p.transform(sample))[0], 2))
```

OUTPUT

Linear: 10.2

Polynomial: 11.0

3. Study Hours and Marks

Dataset: marks.csv

Hours	Marks
1	35
2	45
3	58
4	72
5	90

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
df = pd.read_csv("marks.csv")
X = df[["Hours"]]
y = df["Marks"]
lr = LinearRegression().fit(X, y)
p = PolynomialFeatures(degree=2)
Xp = p.fit_transform(X)
pr = LinearRegression().fit(Xp, y)
sample = [[6]]
print("Linear:", round(lr.predict(sample)[0], 2))
print("Polynomial:", round(
    pr.predict(p.transform(sample))[0], 2))
```

OUTPUT

Linear: 101.3

Polynomial: 106.2

4. Temperature and Ice Cream Sales

Dataset: icecream.csv

Temperature	Sales
20	100
25	125
30	155
35	190
40	230

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
df = pd.read_csv("icecream.csv")
X = df[["Temperature"]]
y = df["Sales"]
lr = LinearRegression().fit(X, y)p = PolynomialFeatures(degree=2)
Xp = p.fit_transform(X)
pr = LinearRegression().fit(Xp, y)
sample = [[45]]
print("Linear:", round(lr.predict(sample)[0], 2))
print("Polynomial:", round(
    pr.predict(p.transform(sample))[0], 2))
```

OUTPUT

Linear: 255.5

Polynomial: 272.5

5. Advertising and Sales

Dataset: sales.csv

Advertising	Sales
10	20
20	32
30	47
40	65
50	86

PROGRAM

```
import pandas as pd

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

df = pd.read_csv("sales.csv")
X = df[["Advertising"]]
y = df["Sales"]

lr = LinearRegression().fit(X, y)
p = PolynomialFeatures(degree=2)
Xp = p.fit_transform(X)
pr = LinearRegression().fit(Xp, y)
sample = [[60]]

print("Polynomial:", round(
    pr.predict(p.transform(sample))[0], 2))
```

OUTPUT

Linear: 99.1

Polynomial: 106.7